

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (IT) Examination

December 2012

IT404 Language Processor

Date: 08.12.2012, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION - I

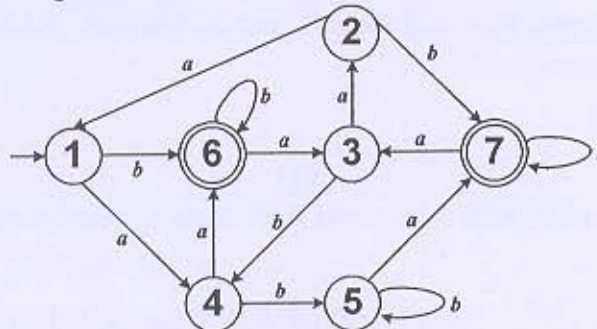
- Q - 1 (a) What is pass? Differentiate between single pass and multiple pass compiler. [03]
(b) Explain two pass assembler in detail. [04]
- Q - 2 (a) Draw NFA- $\wedge$ for following expression and convert it into DFA. Also prepare transition table. [08]
 $(0/1)^*011$

OR

- (a) Define the following terms with example: [08]
(1) synthesized attribute (2) annotated parse tree (3) lexeme (4) handle
- (b) Draw DFA for following RE. [04]
 $(111 + 100)^*0$

OR

- (b) Minimize the following DFA into minimum number of states. [04]



- (c) Differentiate problem-oriented language and procedure-oriented language. [02]
- Q - 3 (a) Explain functionality of preprocessor. [04]
(b) Check following grammar is SLR or not. [06]
$$\begin{aligned} S &\rightarrow aABc \\ A &\rightarrow Abc \mid b \\ B &\rightarrow d \end{aligned}$$
- (c) Check whether the below grammar is Ambiguous or not? [04]
$$\begin{aligned} L &\rightarrow L; L \mid S \\ S &\rightarrow a \end{aligned}$$

OR

- Q - 3 (a) What is language processor? Explain in detail language processing activities. [04]

- (b) Check the following grammar is LL(1) or not. [06]

$$\begin{aligned} S &\rightarrow ACB \mid CbB \mid Ba \\ A &\rightarrow da \mid BC \\ B &\rightarrow g \mid \varepsilon \\ C &\rightarrow h \mid \varepsilon \end{aligned}$$

If it is LL(1) construct the parsing table.

- (c) Perform the left factoring for the following grammar. [04]

$$\begin{aligned} S &\rightarrow iEtS \mid iEtSeS \mid a \\ E &\rightarrow b \end{aligned}$$

SECTION – II

- Q - 4 (a) Match all the items in Table A with appropriate option in Table B. [02]

Table A	Table B
1. Regular expression	1. Syntax analysis
2. Pushdown automata	2. Code generation
3. Dataflow analysis	3. Lexical analysis
4. Register allocation	4. Code Optimization

- (b) What is three address code representation? Write three address code and quadruples for following expression. [05]

$$(2 + A * (B - C / D)) / E$$

- Q - 5 (a) Draw model of LR parser and write LR parser algorithm. [05]

- (b) Write short note on symbol table management. [03]

- (c) Construct canonical LR items and parsing table for following grammar: [06]

$$\begin{aligned} S' &\rightarrow S \\ S &\rightarrow CC \\ C &\rightarrow cC \mid d \end{aligned}$$

OR

- (c) Explain operator precedence parsing. Draw precedence table and perform the parsing for the string : $id * (id \wedge id) - id / id$ [06]

Grammar: $E \rightarrow E+E \mid E-E \mid E * E \mid E / E \mid E \wedge E \mid (E) \mid -E \mid id$

- Q - 6 (a) Write short note on loader. [02]

- (b) Attempt any Three. [12]

1. Dynamic memory allocation.
2. Code optimization techniques.
3. Error recovery strategies.
4. Activation record.
